

Enhancement of Biogas Production Efficiency Using Appropriate Low-Temperature Pretreatments of Municipal Treatment Plants' Excess Sludge

Mehdi Rafiee,^a Elaheh Salehi,^b Khashayar Sharifi,^c Amir H. Mohammadi,^{d,e} Aliasghar Rohani,^c and Morteza Sabeti^a

^aDepartment of Chemical Engineering, Isfahan University of Technology, Isfahan, Iran

^bDepartment of Chemistry, Iran University of Science and Technology, Tehran, Iran

^cDepartment of Refining and Purification Technology, Research Institute of Petroleum Industry (RIPI), Tehran, Iran

^dInstitut de Recherche en Génie Chimique et Pétrolier (IRGCP), Paris Cedex, France

^eDiscipline of Chemical Engineering, School of Engineering, University of KwaZulu-Natal, Durban, South Africa; a.h.m@irgcp.fr; amir_h_mohammadi@yahoo.com (for correspondence)

Published online 22 October 2018 in Wiley Online Library (wileyonlinelibrary.com). DOI 10.1002/ep.13072

Excess sludge in municipal plants normally contains high amounts of organic compounds (chemical oxygen demand [COD] of about 60,000 mg/L) and is capable of producing biogas. Anaerobic digestion is one of current methods to produce biogas. Municipal treatment plants' excess sludge contains high amounts of microbial cells with robust walls which inhibit microorganisms' access to the cells' organic content. This decreases biogas production efficiency as well as prolongs the time of anaerobic digestion process. In this study, a combination of the low-temperature pretreatment techniques was used to increase biogas production efficiency in Isfahan South treatment plant. Through the mentioned techniques, which involve chemical, alkaline, microwave, and mechanical pretreatments, the cells' external structure is decomposed, and microorganisms' access to organic content is facilitated. All the experiments were carried out in two stages. The results of the first stage show that the highest amount of biogas efficiency is 365 mL/g COD for the sample under alkaline pretreatment by NaOH at 20 °C and pH = 12. The results of the second stage demonstrate that the highest rates of biogas production efficiency are obtained in samples under chemical pretreatment by KOH on secondary sludge at an ambient temperature (22 °C), and under 2400 H₂O₂ oxidation, respectively. © 2018 American Institute of Chemical Engineers Environ Prog, 38:e13072, 2019

Keywords: low-temperature pretreatments, anaerobic digestion, biogas production, excess sludge, gas production

INTRODUCTION

Energy plays a fundamental role in the life of societies' industrial economy. As natural sources are finite and terminable, more attempts are required to develop and promote novel renewable energy resources [1]. To this end, loads of researches have been conducted in this field. The many benefits associated with biogas as a renewable energy resource include reducing dependency on fossil fuels and lowering the

emission of pollutant gases from industrial production. Not only does biogas provide a good source of generating energy, but it also contributes to generating agricultural manure, promotes general health, and controls diseases [2]. Anaerobic fermentation process, which results in methane and carbon dioxide, is one of several desirable techniques used in generating environmentally clean energies. Many studies have used various substrates to generate biogas, among which are municipal treatment plants' excess sludge, wastewater collected from different industrial sections, and lignocellulosic materials [3,4].

Anaerobic digestion is a treatment process used to stabilize sludge in many water treatment plants. It has many advantages such as reducing sludge mass, generating methane, and improving dewatering properties of the refined sludge. However, the low speed of sludge decomposition leads to solids retention times up to 20–30 days in conventional mesophilic digesters (at 37 °C), which demand much more space for the facilities due to the huge digesters [5]. Along with the anaerobic digestion systems growth, a variety of techniques are utilized to promote the anaerobic digestive efficiency, each of which somehow either increases the biogas production rate or decreases the hydraulic retention time of the digesters. Various pretreatments of the activated sludge to enhance biogas production efficiency in anaerobic digestion systems include:

1. chemical pretreatment [6–11],
2. alkaline pretreatment [12–18],
3. thermal pretreatment [5,6,9,10,13–15,19–27],
4. thermal-chemical pretreatment [6,9,10,27],
5. mechanical (ultrasonic) pretreatment [12,20,28–31], and
6. microwave pretreatment [2,32].

The advantages and disadvantages of using each pretreatment method were gathered and analytically interpreted by Karimi and Taherzadeh [33]. In 2009, Lin *et al.* reported an increase of 183% in biogas production using the alkaline pretreatment on paper sludge [16]. Lin *et al.* [17] found an increase of 53% in hydrogen production potential using the alkaline pretreatments using NaOH. Chi *et al.* [2] imposed alkaline and

microwave pretreatments on the sludge from a treatment plant and found a decrease in H₂S within the yielded biogas. Moreover, an increase in the generated methane and solubility ratio was observed. Some other reached similar conclusions by adding other alkaline materials such as KOH and chemical materials like H₂O₂ [6,12,13]. In their study, Devlin *et al.* [7] concluded that the digestion period decreased and that methane production efficiency increased by 14% when doing an acidic pretreatment by HCl. Bougrier *et al.* [25] investigated the effect of high-temperature alkaline pretreatment on biogas production and recorded an 80% increase in the production efficiency. Rani *et al.* added NaOH at a temperature of 60 °C and observed an increase of 51% in biogas production efficiency [14]. The effects of pretreatment process with 1–2% w/v NaOH at high temperatures were investigated economically by Safari *et al.* [34]. Finally, the researchers indicated that their pretreatment could tangibly decrease the manufacturing costs of ethanol production from pinewood.

Many researchers investigated the effect of thermal pretreatment on the biogas production rate, and other key parameters. For instance, Ferrer *et al.* [5] developed a thermophilic lab-scale digester to test the effect of pretreatment temperature and time on methane production in mixture of primary and secondary waste sludge. They showed that when the waste sludge was initially heated for 9 h at 70 °C, the volume of methane produced by the thermophilic anaerobic digestion could increase by 30% in their lab-scale reactor. In a similar study, Ge *et al.* [21] reported an increase of 25% in the biogas production rate using activated sludge for 48 h at 60 °C. Liu *et al.* [35] examined the effect of thermal pretreatment of sludge at 175 °C for 60 min. They observed a decrease in sludge viscosity, an increase in its dewatering property, and a two-time increase in the methane production rate. Park and Ahn [32] investigated the optimal point of microwave pretreatment. The results showed an increase of 14% in solubility ratio and an increase of 809 mL/g volatile solids (VS) in methane efficiency.

For a long time, there has been a growing attention toward the effect of ultrasonic pretreatment on the biogas production rate. To name a few, Dhar *et al.* [29] exerted ultrasonic waves on the activated sludge for 30 min under varying powers, yielding an increase of 30% in the biogas production rate. Similarly, Bougrier and Kim reported an increase in the biogas production rate, solubility, and chemical oxygen demand (COD) [12,27].

In the present study, a variety of experiments were conducted on samples of excess sludge taken from the municipal treatment plant, located in south of Isfahan, to solve problems pertaining to the biogas production efficiency by the digesters, as demanded by the industrial section. The mentioned excess sludge contains enormous amounts of substrates along with organic compounds. The purpose of this study was to breakdown the related cellular structure and to release the cells' organic ingredients with the help of an appropriate pretreatment technique. To this end, a variety of low temperature pretreatment techniques including chemical, ultrasonic, and microwave are utilized. Finally, the rate of biogas production is examined in terms of temperature and oxidizing concentration.

METHODOLOGY AND EXPERIMENT

Primary Materials

The required excess activated sludge containing high oxygen demand value was collected from Isfahan municipal treatment plant in two phases of the mixed (or primary) sludge and secondary sludge (as shown in Fig. 1). After preparing the required substrate, it was immediately conserved at 4 °C in a refrigerator. During sampling in each phase, the microbial mixture (inoculum) was again prepared from the treatment plant

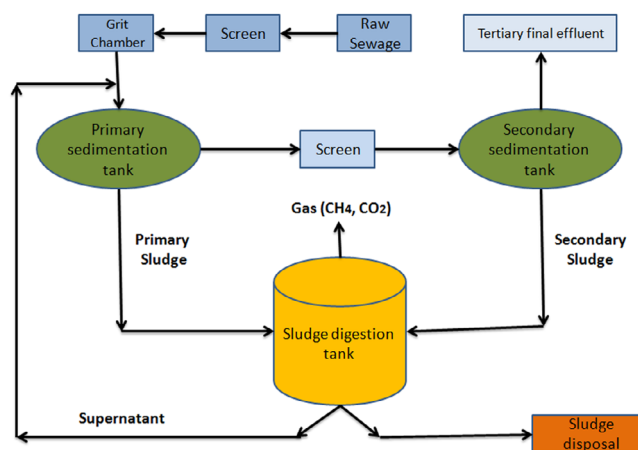


Figure 1. Schematic of Isfahan's sewage treatment plant which was used in this work. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

as needed. It should be also noted that the mentioned mixture was prepared from the anaerobic part of the treatment plant. Table 1 demonstrates the initial characteristics such as the total amount of solids as well as that of VS of the two mentioned activated sludge samples.

Measuring Methods

To examine and analyze how the process of anaerobic fermentation works, it is necessary to measure the key properties of the sludge including its COD, total solids (TS) and VS amounts, and also the produced biogas rate from the feeds.

COD Measuring Method

The D-5220 closed reflux, colorimetric method was used to measure the COD of the materials [36].

Measuring TS and VS Amounts

One method to measure the amount of organic materials within the sample is measuring the amount of VS. The yielded biogas from the Lignocellulosic compounds is expressed in terms of VS unit. In this study, TS and VS percentages were measured by evaporating the sample inside an oven and furnace. To this end, a certain amount of the sample was poured inside a crucible. Then, it was dried inside an oven at 150 °C, placed in an electrical furnace, dried, and burned at 550 °C for 1 h. TS and VS percentages are calculated through the following relations:

$$TS = \frac{W_3 - W_1}{W_2 - W_1} \times 100\% \quad (1)$$

$$VS = \frac{W_3 - W_4}{W_2 - W_1} \times 100\% \quad (2)$$

According to the above relations, W_1 is the empty crucible weight, W_2 is the crucible and sample weight, W_3 is the crucible and dried sample, and W_4 is the crucible and burnt sample.

Measuring and Analyzing Produced Biogas

The produced biogas and methane; and carbon dioxide percentages were analyzed by a gas chromatographer equipped with a thermal conductivity detector. It is equipped with Peak ABC software, which analyzes the figures resulted

Table 1. Characteristics of the samples and inoculums used in this study.

	TS (%)	VS (%)	TS concentration (g/L)	VS concentration (g/L)
Mixed sludge	6.781	4.558	61.000	41.500
Secondary sludge	7.713	4.362	69.050	39.050
Inoculum 1	6.744	3.083	58.950	26.950
Inoculum 2	6.744	3.083	58.950	26.950

Table 2. The pretreatment processes over the mixed sludge samples prior to the anabolic digestion process.

Sample number	Pretreatment method	Description
1	No pretreatment	-
2	Ultrasonic 20	At 20 °C and for 30 min
3	Microwave 20	At 20 °C and for 5 min
4	Alkaline 20	At 20 °C and for 6 h
5	Ultrasonic 40	At 40 °C and for 30 min
6	Microwave 40	At 40 °C and for 5 min
7	Alkaline 40	At 40 °C and for 6 h

from the samples injection. Nitrogen was utilized as a carrier gas with a 20 mL/min flow rate. The operational conditions were regulated as follows: column temperature: 50 °C; injector temperature: 100 °C; and detector temperature: 150 °C.

Initially, standard peaks of methane, CO₂, and hydrogen were determined according to certain conditions and rates of the gas chromatography (GC) carrier gas. To this end, the required samples were taken from methane, CO₂, and hydrogen cylinders using specific impenetrable pockets. Then, varying volumes of the produced pure gases were injected into the GC using a 100 microliter controlling syringe, and the yielded peaks were recorded as standard values. This process was repeated three times for each of the gases, and finally, average values for under peak area and injected gas volume were computed. Moreover, the time of gas emission was detected by the detector. Biogas percentage composition was calculated by comparing the obtained peaks from the produced gas with the standard ones.

After each sampling time, to take the pressure inside the reactors to the ambient pressure, the produced gas was ejected from the reactors by the syringe needle. Then, the reactors were placed inside an incubator to reproduce the biogas. To remove the produced biogas, after inserting the syringe needle into the reactors, the reactors were put under water to observe the exited biogas bubbles. When the last bubble was observed, the pressure inside the reactors was assured to be equal to the ambient pressure.

The Procedure of the Experiment

Pretreatment Process

To accelerate the hydrolysis process and decompose the external structure of the microbial cells with the aim of facilitating bacteria's access to the organic materials, chemical, ultrasonic, and microwave pretreatments were utilized at two different temperatures. Tables 2 and 3 show the methods used during pretreatment processes in this study. The results were then compared with one another to identify the more beneficiary processes. The types of utilized alkaline and selected pH value were chosen according to the optimal amounts available in scientific papers. The experiments were carried out based on what the industry field (Isfahan south treatment plant)

Table 3. The pretreatment processes over the secondary sludge samples prior to the anabolic digestion process.

Sample number	Pretreatment methods	Description
1	No pretreatment	-
2	KOH	At pH 12 and for 6 h.
3	NaOH	At pH 12 and for 6 h.
4	HCl	At pH 2 and for 6 h.
5	H ₂ O ₂ 150 (mg/L)	At pH 6 and for 6 h.
6	H ₂ O ₂ 600 (mg/L)	At pH 6 and for 6 h.
7	H ₂ O ₂ 1200 (mg/L)	At pH 6 and for 6 h.
8	H ₂ O ₂ 2400 (mg/L)	At pH 6 and for 6 h.

demanding in order to solve the problems pertaining to the digesters of the treatment plant. To avoid repeating some experiments, in some cases, determining the optimal amounts was based on the results taken from available scientific papers.

First, 200 mL of mixed sludge diluted by deionized water with a ratio of 50:50 was poured into some bottles (Erlenmeyer Flasks). Then, some of them were exposed to ultrasonic waves for 30 min with a power of 20 kHz and microwave for 5 min with a power of 300 W. An alkaline additive (NaOH) was introduced into the other samples until their pH reached 12. It was also tried to keep the samples at two temperatures of 20 °C and 40 °C during the pretreatment processes.

Second, similar to the mixed sludge samples, some bottles with the secondary diluted sludge samples were prepared. Then, sodium hydroxide, potassium hydroxide, hydrochloric acid, and hydrogen peroxide with varying concentrations separately were added to the bottles. A delicate attention was paid to keep the pH values at 12 for NaOH/KOH, 2 for HCl, and 6 for H₂O₂ pretreatment processes. After 6 h, all Erlenmeyer flasks reached a neutral pH through adding 6 molar of either sodium hydroxide or hydrochloric acid, and became ready to be injected into biogas bottles. During the whole additive pretreatment process which lasted for 6 h, magnetic stirring was applied to all samples. The pH-meter probe was used to regularly measure and monitor ambient temperature, the whole-sample temperature as well as acid and alkaline samples' pH values.

The Production of Biogas in a Batch Reactor

In this study, glass bottles of 118 mL were used as batch reactors. A volume of 20 mL of the inoculum and 10 mL of the sludge, which was first diluted by deionized water with a ratio of 50:50 and then pretreated, were poured into each of the bottles. The bottles were tightly closed by rubber corks and aluminum caps. Then, nitrogen gas was injected into the bottles for 2 min by a syringe. At the same time, the air within the

bottles was emptied by another syringe in order to make the system as an anaerobic system. To make sure that the bottles were vacuumed, hot glue was used to fasten the bottles' caps in every testing time. This batch system, known as Hansen

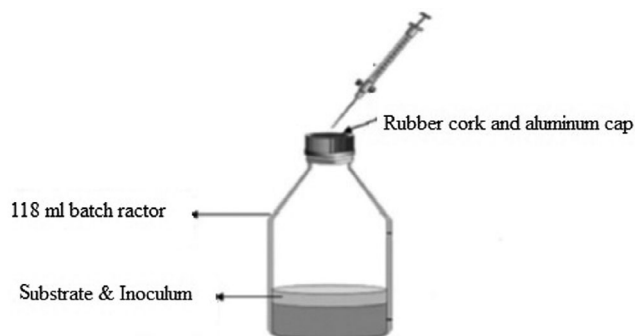


Figure 2. Schematic of a batch bioreactor used in this study.

method, has been used in various experiments as a batch reactor [37].

All the mentioned tests were separately conducted two times so as to ensure the accuracy of results. To see the effect of pretreatment operation, a sample with similar conditions and no pretreatment was also examined. Figure 2 shows a schematic for the employed biogas reactor, and Figure 3 reveals a schematic for reactors initiation stages. Finally, the results of injecting the yielded biogas into the GC were stored and utilized for the calculation of biogas compound percentage as well as methane and carbon dioxide volumes.

RESULTS AND DISCUSSION

To examine the effect of pretreatment process on the total dissolved organic solids present in the sludge, the related COD value was measured after each process. The samples' COD values in each process were calculated relative to the untreated sample (the crude sludge). The results are demonstrated in Figures 4 and 5 for the mixed and secondary sludge, respectively.

According to Figure 4, the pretreatment process led to an increase in the mixed sludge COD level, which is a criterion of the amount of organic compounds within it. In fact, this



Figure 3. Procedure of biogas production using batch reactors: (a) adding inoculum mixtures, (b) adding substrates, (c) putting the caps on the reactors, (d) driving oxygen away using nitrogen gas, (e) discharging extra nitrogen and setting the inside reactors' pressure at 1 atm, and (f) putting the reactors inside an incubator. [Color figure can be viewed at wileyonlinelibrary.com]

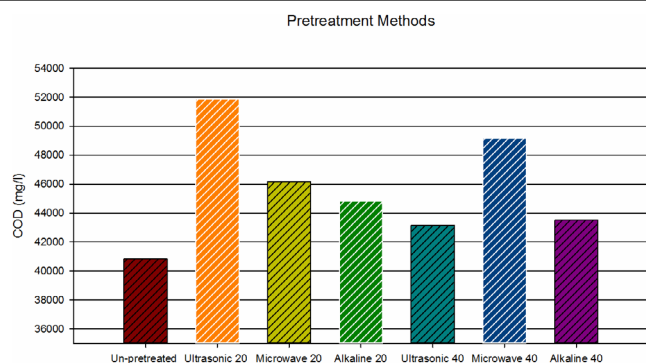


Figure 4. The results of measuring the mixed sludge COD through chemical, ultrasonic, and microwave processes at 20 °C and 40 °C. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

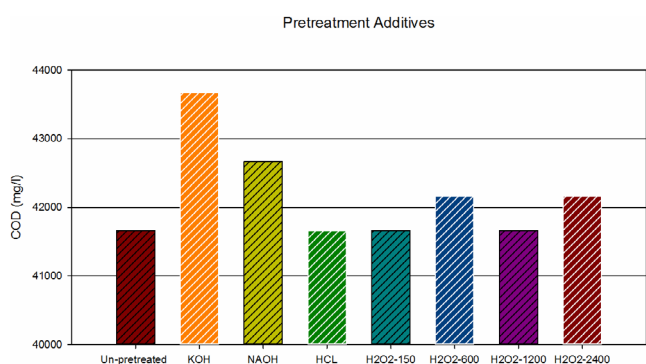


Figure 5. The results of measuring the secondary sludge COD through chemical pretreatment at 20 °C. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

process results in releasing the organic matters within the cellular structures surrounded by polymers. The level of increase is varying depending on each pretreatment method and is followed by different mechanisms of each method affecting cellular structures present in the sludge. The highest increase in COD concerns with the ultrasonic pretreatment at 20 °C, which equals to 27%, and the lowest COD amount pertains to the ultrasonic pretreatment at 40 °C, which equals to 5.7%. For the other methods, an increase of more than 6% in COD amount is observed. The results of the chemical pretreatment on the secondary sludge show an increase of COD value in all the cases, with the highest increase value being related to alkaline pretreatment using KOH, which equals to 5%.

Figures 6 and 7 demonstrate the results of COD variations after the biogas production period, respectively. As Figure 6 reveals, after the biogas production period, the sample COD amount experienced a decrease in all bottles pretreated with different methods. The highest decrease rate (57%) was yielded from the bottle pretreated by the alkaline method at 20 °C. After that, the bottles pretreated by ultrasonic method at 20 °C, microwave method at 40 °C, and microwave method at 20 °C showed downfalls of 55%, 53%, and 49%, respectively. As per the figure, a decrease of 31% in COD level is observed for the unpretreated sample. The amount of COD decrease can be a criterion of converting the soluble organic matters present in the sample into carbon biogas.

The values of the secondary sludge COD after biogas production period are presented in Figure 7. Here, the sample pretreated by H₂O₂ with the concentration of 2400 mg/L has the highest COD reduction, which is equal to 34%. As the values of COD in the column chart decreased after the digestion process, biogas is expected to be produced based on the sample's COD consumption. The results of the biogas production from the secondary sludge prove this claim.

The biogas production inside the reactors using the chemical pretreatment, ultrasonic, and microwave pretreatments at 20 °C and 40 °C lasted for 22 days for the first phase (using the mixed sludge) and 28 days for the second phase (using the secondary sludge). In the last 2 days, in practice, the biogas production rate reached zero. The rate of the biogas

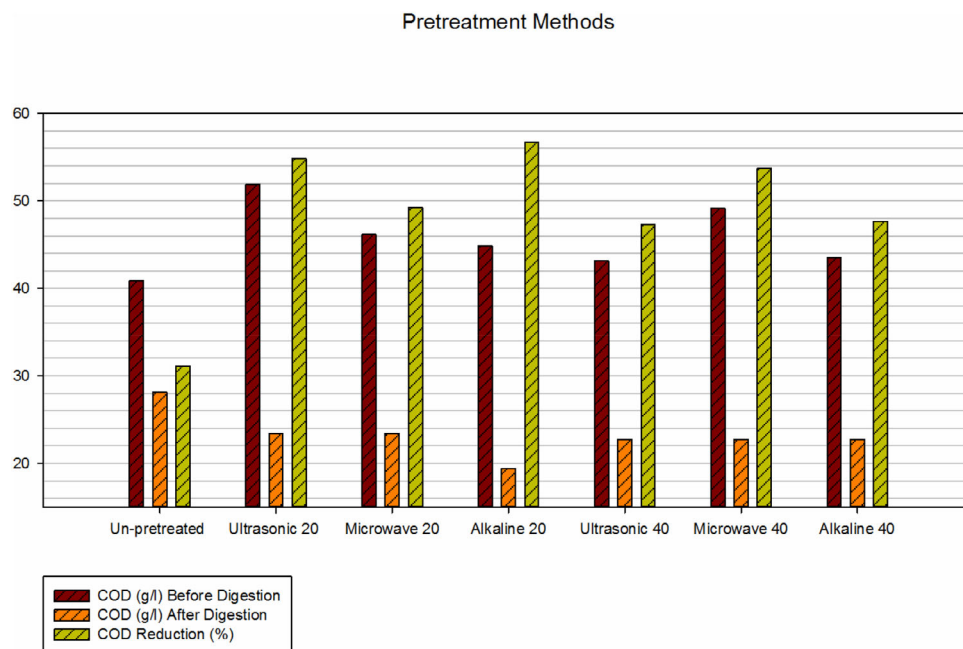


Figure 6. The amount of COD after and before the biogas production period for the mixed sludge. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

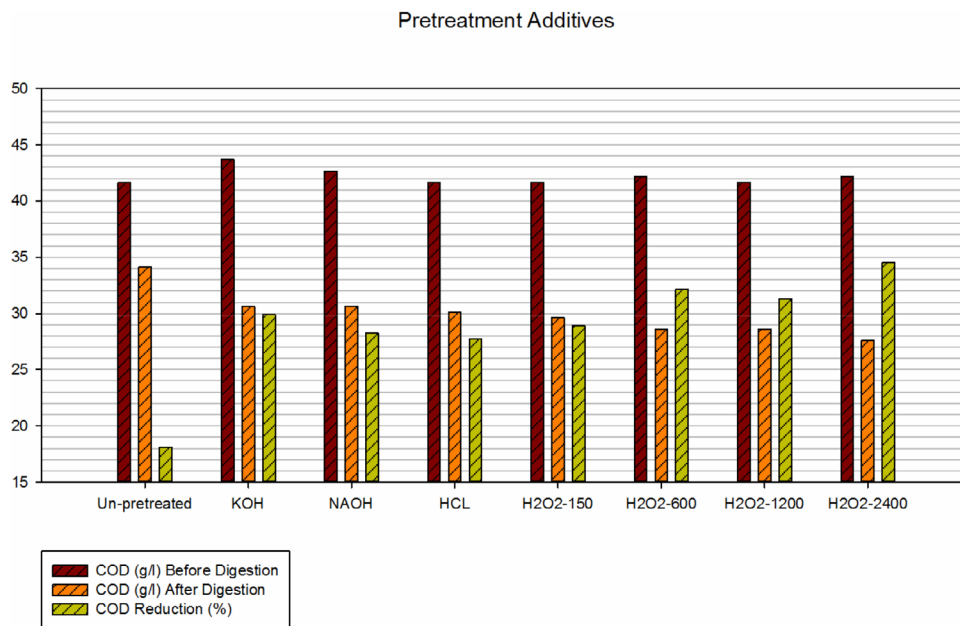


Figure 7. The amount of COD after and before the biogas production period for the secondary sludge. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

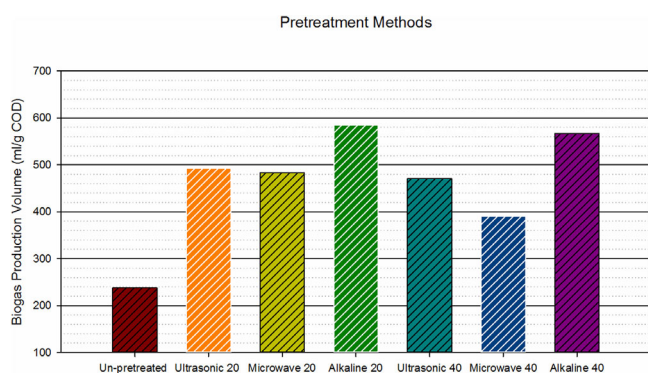


Figure 8. The amount of produced biogas using different pretreatments on the mixed sludge. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

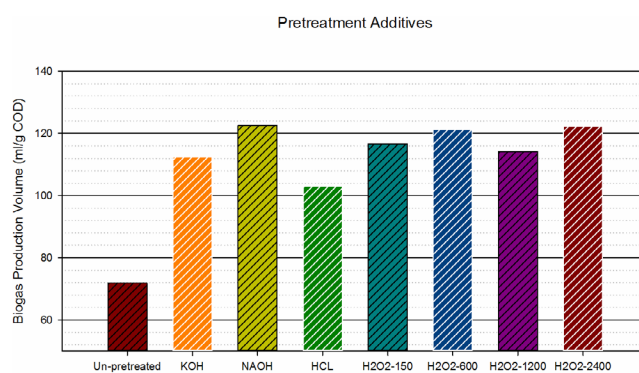


Figure 9. The amount of produced biogas using different pretreatments on the secondary sludge. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

production for both phases varied due to the different substrates as well as different pretreatment compounds used in the biogas production procedure in the second phase. However, the levels of biogas production in both periods came in agreement with those reported by scientific papers. The results are provided in Figures 8 and 9.

Concisely, Figure 10 presents the cumulative yielded methane versus time for the mixed sludge pretreated at 20 °C. The highest level of produced methane is 365 mL/g COD, showing a 3.5 time increase compared with the unpretreated yield (160 mL/g COD). The pretreated samples using ultrasonic and microwave methods are in the second and third ranks with 345 and 274 mL/g COD, respectively.

According to Figure 11, the highest amount of yielded biogas has occurred in the days 8 to 10. The produced biogas rate in the 22th day has reached zero.

Figure 12 demonstrates a schematic for the mixed sludge pretreated at 40 °C. The rate of produced biogas pretreated at 40 °C is less than that pretreated at 20 °C. Moreover, the highest level of produced methane is 352 mL/g COD for the

alkaline pretreatment, which shows a three-time increase compared with the unpretreated sample.

The values of biogas yielded from the mixed sludge by different techniques at 40 °C are presented in Figure 13. Here, a two-time increase is observed in methane produced from pretreated samples compared with unpretreated ones. Regarding the mixed sludge, it is observed that the highest methane percentages being under pretreatment by NaOH at 20 °C and ultrasonic at 20 °C are 365 and 345 mL/g COD, respectively, because of the high amount of pretreated organic content available for the microorganisms to digest. The results also show that the samples under alkaline pretreatment at 20 °C and 40 °C and the samples under ultrasonic pretreatment at 20 °C, compared with the pure samples, had the highest increase in COD. Moreover, the highest COD reduction was yielded from the samples under the ultrasonic and alkaline pretreatments at 20 °C. The conversion of more organic content available in the samples into methane, which led to a more methane percentage than CO₂, has occurred in those samples under the alkaline pretreatment at 20 °C and 40 °C and ultrasonic pretreatment at 20 °C.

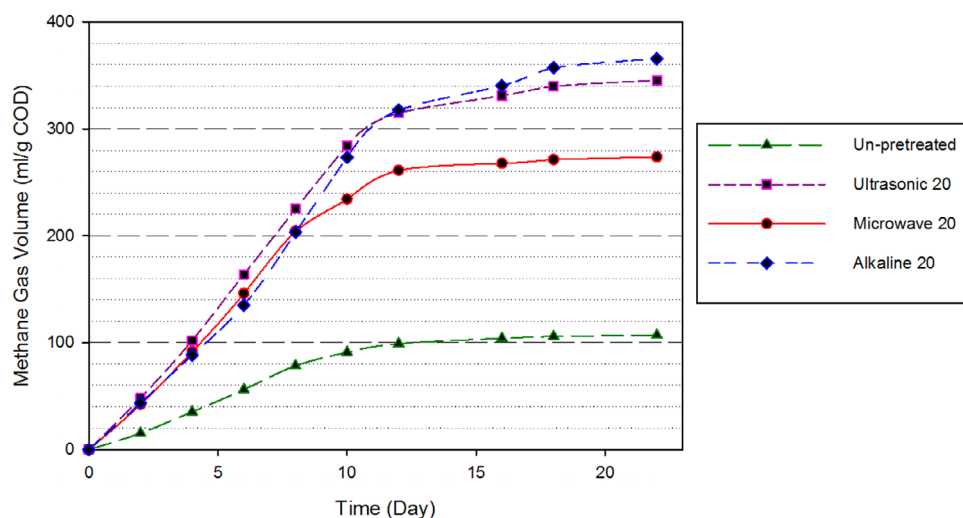


Figure 10. The cumulative methane yielded from the mixed sludge pretreated at 20 °C. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ep.13072)]

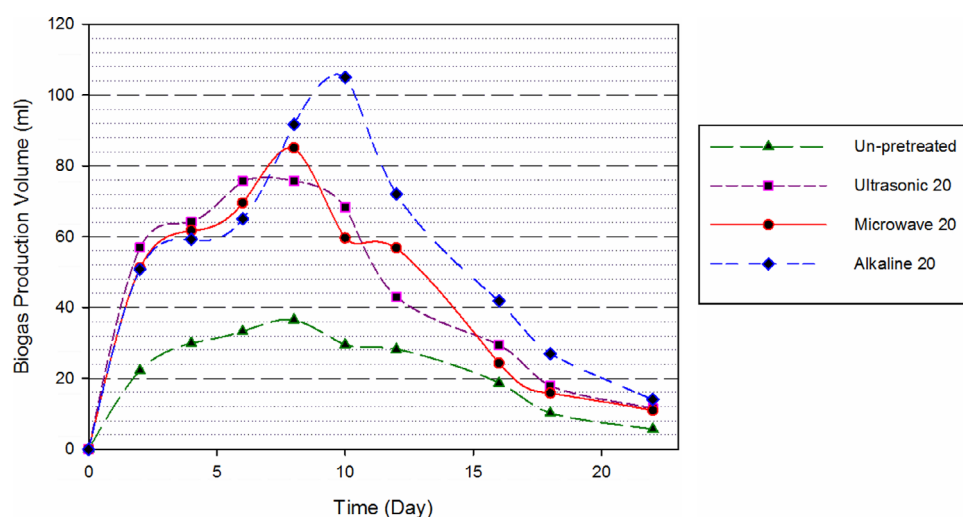


Figure 11. The amount of produced biogas based on time period during 22 days of pretreating the mixed sludge at 20 °C. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ep.13072)]

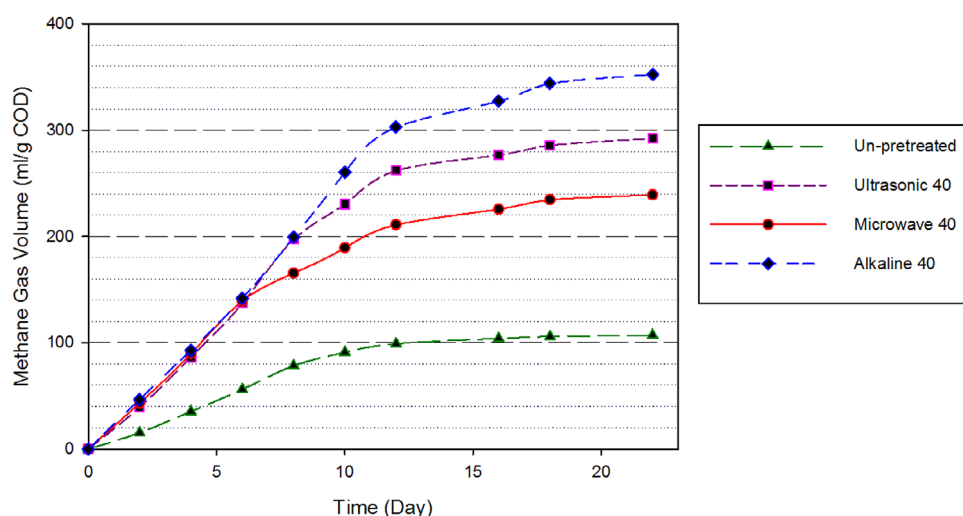


Figure 12. The cumulative methane yielded from the mixed sludge pretreated at 40 °C. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ep.13072)]

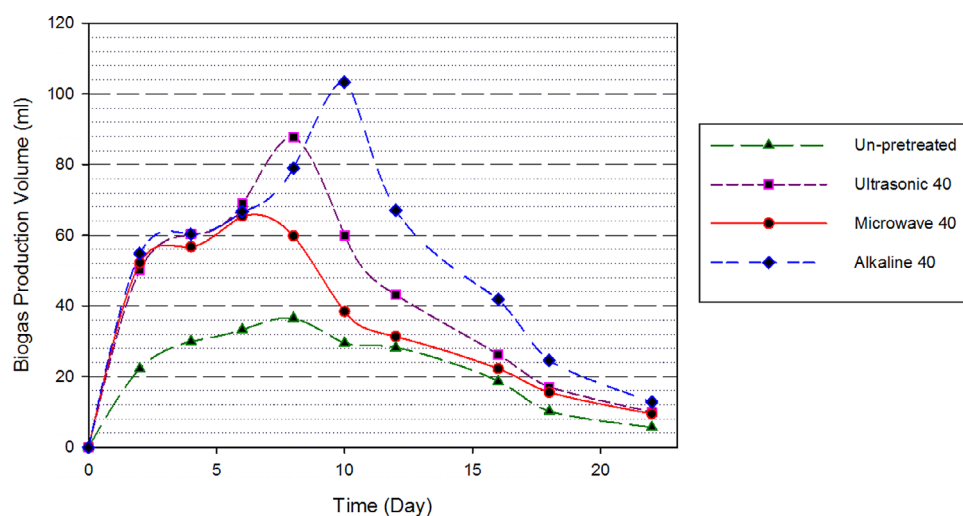


Figure 13. The amount of produced biogas based on time period during 22 days of pretreating the mixed sludge at 40 °C. [Color figure can be viewed at wileyonlinelibrary.com]

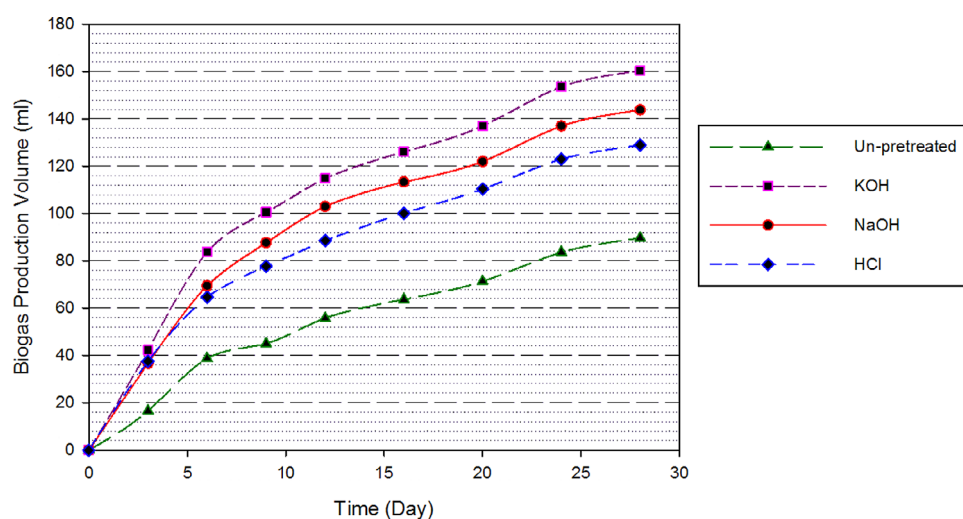


Figure 14. The cumulative biogas production rates from the secondary sludge samples pretreated by different chemical additives at 20 °C. [Color figure can be viewed at wileyonlinelibrary.com]

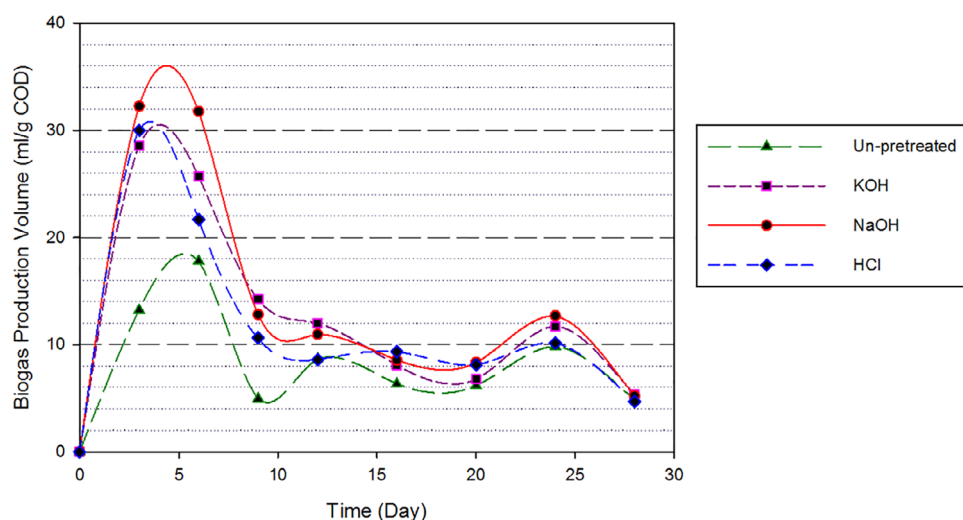


Figure 15. The rates of biogas production from the secondary sludge samples pretreated by different chemical additives at 20 °C within 28 days. [Color figure can be viewed at wileyonlinelibrary.com]

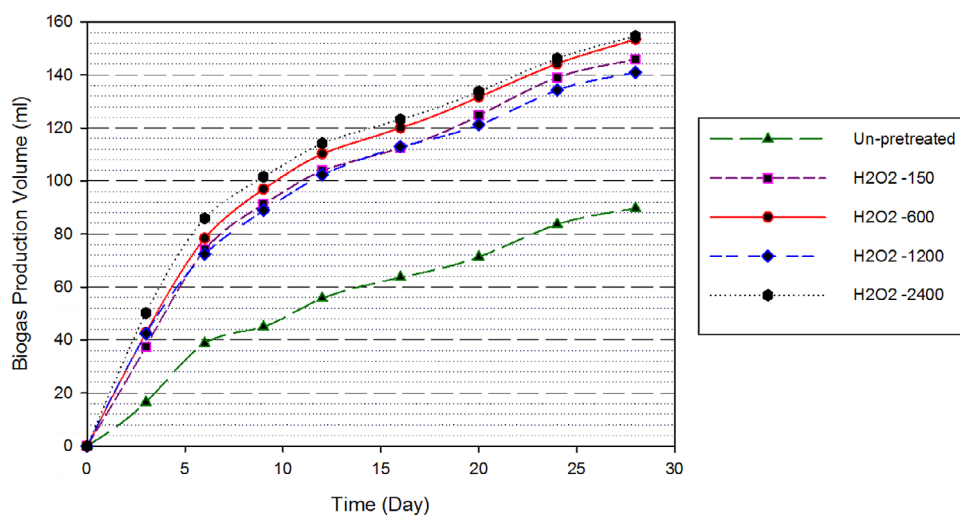


Figure 16. The cumulative biogas production rates from the secondary sludge samples pretreated by a variety of H_2O_2 concentrations at 20°C . [Color figure can be viewed at wileyonlinelibrary.com]

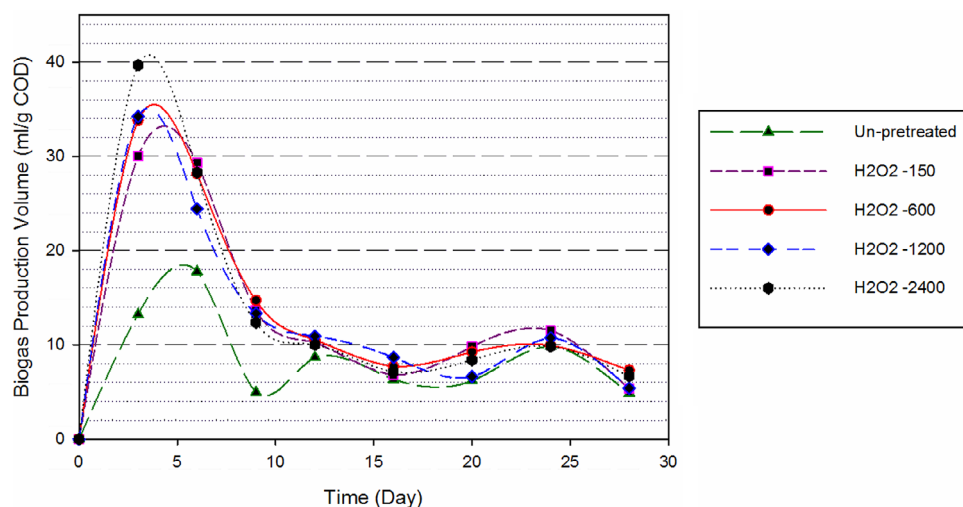


Figure 17. The rates of biogas production from the secondary sludge samples pretreated by a variety of H_2O_2 concentrations at 20°C . [Color figure can be viewed at wileyonlinelibrary.com]

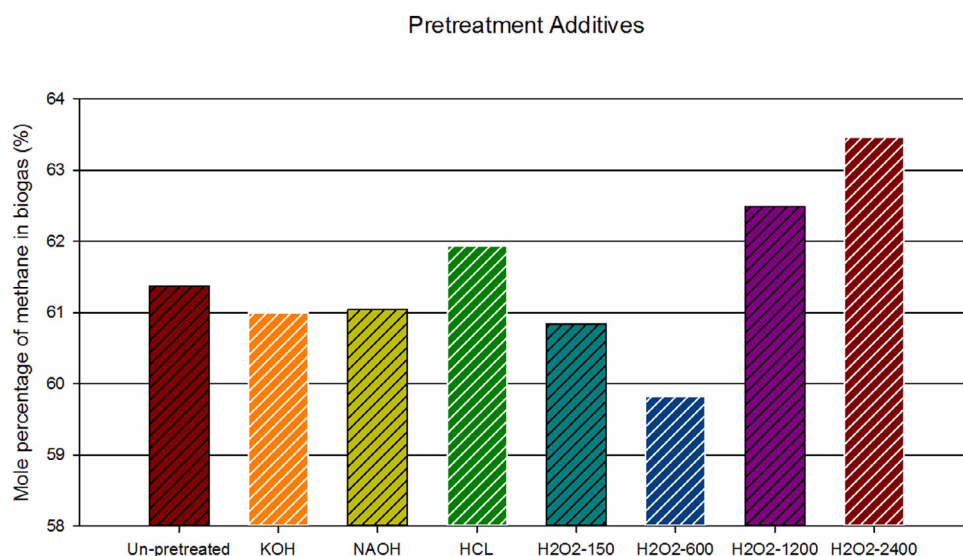


Figure 18. The molar percentage of methane gas existing in the biogas during digestion of the secondary sludge employing the defined chemical additives. [Color figure can be viewed at wileyonlinelibrary.com]

The results of measuring COD well justify higher efficiency of biogas production in the alkaline samples.

In this part, the results of experiments regarding the secondary sludge are presented. The cumulative diagrams (Figure 14) related to the biogas produced from the samples pretreated by alkaline and acid methods demonstrate that alkaline pretreatment by KOH is more effective than that by NaOH. The value of biogas produced from the secondary sludge pretreated by KOH equals 160 mL, showing a 78% increase compared with the amount of biogas produced from the untreated sludge. Then, those samples pretreated by NaOH and HCl are in the second and third ranks, with an increase of 60% and 43%, respectively.

Figure 15 presents the results of biogas daily produced from the secondary sludge pretreated by acid and alkaline methods with pH of 12 and 2, respectively. It is observed that the highest rate of biogas production is related to the sample pretreated by NaOH in days 4 to 7.

Figure 16 provides the results of biogas produced from the secondary sludge chemically pretreated by H_2O_2 under varying concentrations. The amounts of biogas produced from the pretreated samples are more than that from the untreated sludge, and the sample pretreated by H_2O_2 with concentration of 2400 mg/L has the highest amount (155 mL).

Figure 17 presents the results of biogas daily produced from the secondary sludge pretreated by H_2O_2 under varying concentrations. In this case, the highest amount of biogas produced from the secondary sludge is 40 mL/g COD in the 4th day. Moreover, the highest amount of pretreated sample by H_2O_2 has occurred when the concentration is 2400 mg/L.

Figure 18 reports the amount of biogas yielded from the secondary sludge pretreated by different methods based on percentage. The value of methane available in the biogas produced from the untreated sludge equals 61%, while concerning the sample pretreated by H_2O_2 with the concentration of 2400 mg/L, the value has reached about 64%. The highest production efficiency of methane yielded from the pretreated secondary sludge is related to the sample pretreated by H_2O_2 with the concentration of 2400 mg/L and the sample under alkaline pretreatment by KOH with a pH of 12. The increase rates for each case compared with the untreated sludge are 69% and 70%, respectively.

The highest COD reduction through the anaerobic digestion of the secondary sludge under different pretreatment methods concerns the sample pretreated by H_2O_2 with the concentration of 2400 mg/L, which equals 34%.

The highest increase level of COD concerns the sample under alkaline pretreatment by KOH. The results of measuring COD well justify higher efficiency of biogas production in alkaline samples.

CONCLUSION

Two substrates containing high amounts of organic compounds were taken from Isfahan municipal treatment plants to examine the effect of the low-temperature pretreatment on the biogas production efficiency. The results show that the mixed sludge being under alkaline pretreatment by NaOH had the highest amount of efficiency (365 mL CH_4 /g COD) with a pH of 12 at 20 °C. Then, the sample pretreated by ultrasonic method had the efficiency of 345 mL/g COD at 20 °C.

For the secondary sludge samples, the highest values of biogas production are related to the sample pretreated by H_2O_2 with the concentration of 2400 mg/L and the sample pretreated by KOH with a pH of 12 with the efficiency percentages of 72 and 64, respectively.

The results also proved that in all cases, pretreatment process caused an increase in the COD level, with the highest increase occurring at 20 °C and an efficiency of 27% for the sample pretreated by the ultrasonic method.

The other results of this study are concerned with the relationship between the decrease of COD after biogas production period with biogas production increase as well as the amount of methane available in the yielded biogas. In addition, the results indicated that the alkaline technique had the highest efficiency and potential among other pretreatment techniques. It should be mentioned that some other parameters additionally affect the amount of biogas production which were not in scope of this study.

LITERATURE CITED

- Kim, J., Park, C., Kim, T.H., Lee, M., Kim, S., Kim, S.W., & Lee, J. (2003). Effects of various pretreatments for enhanced anaerobic digestion with waste activated sludge, *Journal of Bioscience and Bioengineering*, 95(3), 271–275.
- Chi, Y., Li, Y., Fei, X., Wang, S., & Yuan, H. (2011). Enhancement of thermophilic anaerobic digestion of thickened waste activated sludge by combined microwave and alkaline pretreatment, *Journal of Environmental Sciences*, 23(8), 1257–1265.
- Carrère, H., Sialve, B., & Bernet, N. (2009). Improving pig manure conversion into biogas by thermal and thermo-chemical pretreatments, *Bioresource Technology*, 100(15), 3690–3694.
- Kana, E. G., Oloke, J., Lateef, A., & Adesiyun, M. (2012). Modeling and optimization of biogas production on saw dust and other co-substrates using artificial neural network and genetic algorithm, *Renewable Energy*, 46, 276–281.
- Ferrer, I., Ponsá, S., Vázquez, F., & Font, X. (2008). Increasing biogas production by thermal (70 °C) sludge pre-treatment prior to thermophilic anaerobic digestion, *Biochemical Engineering Journal*, 42(2), 186–192.
- Cacho Rivero, J.A., Madhavan, N., Suidan, M.T., Ginestet, P., & Audic, J.M. (2006). Enhancement of anaerobic digestion of excess municipal sludge with thermal and/or oxidative treatment, *Journal of Environmental Engineering*, 132(6), 638–644.
- Devlin, D., Esteves, S., Dinsdale, R., & Guwy, A. (2011). The effect of acid pretreatment on the anaerobic digestion and dewatering of waste activated sludge, *Bioresource Technology*, 102(5), 4076–4082.
- Dhar, B.R., Youssef, E., Nakhla, G., & Ray, M.B. (2011). Pretreatment of municipal waste activated sludge for volatile sulfur compounds control in anaerobic digestion, *Bioresource Technology*, 102(4), 3776–3782.
- Wood, N., Tran, H., & Master, E. (2009). Pretreatment of pulp mill secondary sludge for high-rate anaerobic conversion to biogas, *Bioresource Technology*, 100(23), 5729–5735.
- Paul, E., Camacho, P., Sperandio, M., & Ginestet, P. (2006). Technical and economical evaluation of a thermal, and two oxidative techniques for the reduction of excess sludge production, *Process Safety and Environmental Protection*, 84(4), 247–252.
- Chu, L., Yan, S., Xing, X.H., Sun, X., & Jurcik, B. (2009). Progress and perspectives of sludge ozonation as a powerful pretreatment method for minimization of excess sludge production, *Water Research*, 43(7), 1811–1822.
- Kim, D.-H., Jeong, E., Oh, S.E., & Shin, H.S. (2010). Combined (alkaline+ ultrasonic) pretreatment effect on sewage sludge disintegration, *Water Research*, 44(10), 3093–3100.
- Valo, A., Carrere, H., & Delgenes, J.P. (2004). Thermal, chemical and thermo-chemical pre-treatment of waste activated sludge for anaerobic digestion, *Journal of Chemical Technology and Biotechnology*, 79(11), 1197–1203.
- Rani, R. U., Kumar, S. A., Kaliappan, S., Yeom, I.-T., & Banu, J. R. (2012). Low temperature thermo-chemical pretreatment of dairy waste activated sludge for anaerobic digestion process, *Bioresource Technology*, 103(1), 415–424.

15. Nielsen, H.B., Thygesen, A., Thomsen, A.B., & Schmidt, J. E. (2011). Anaerobic digestion of waste activated sludge—Comparison of thermal pretreatments with thermal inter-stage treatments, *Journal of Chemical Technology and Biotechnology*, 86(2), 238–245.
16. Lin, Y., Wang, D., Wu, S., & Wang, C. (2009). Alkali pretreatment enhances biogas production in the anaerobic digestion of pulp and paper sludge, *Journal of Hazardous Materials*, 170(1), 366–373.
17. Lin, Y., Liang, J., Wu, S., & Wang, B. (2013). Was pretreatment beneficial for more biogas in any process? Chemical pretreatment effect on hydrogen–methane co-production in a two-stage process, *Journal of Industrial and Engineering Chemistry*, 19(1), 316–321.
18. Shetty, D.J., Kshirsagar, P., Tapadia-Maheshwari, S., Lanjekar, V., Singh, S.K., & Dhakephalkar, P.K. (2017). Alkali pretreatment at ambient temperature: A promising method to enhance biomethanation of rice straw, *Biore-source Technology*, 226, 80–88.
19. Qiao, W., Yan, X., Ye, J., Sun, Y., Wang, W., & Zhang, Z. (2011). Evaluation of biogas production from different biomass wastes with/without hydrothermal pretreatment, *Renewable Energy*, 36(12), 3313–3318.
20. Kim, D.-J., & Youn, Y. (2011). Characteristics of sludge hydrolysis by ultrasound and thermal pretreatment at low temperature, *Korean Journal of Chemical Engineering*, 28(9), 1876–1881.
21. Ge, H., Jensen, P.D., & Batstone, D.J. (2010). Pre-treatment mechanisms during thermophilic–mesophilic temperature phased anaerobic digestion of primary sludge, *Water Research*, 44(1), 123–130.
22. Gavala, H.N., Yenal, U., Skiadas, I.V., Westermann, P., & Ahring, B.K. (2003). Mesophilic and thermophilic anaerobic digestion of primary and secondary sludge. Effect of pre-treatment at elevated temperature, *Water Research*, 37(19), 4561–4572.
23. Climent, M., Ferrer, I., Baeza, M.M., Artola, A., Vázquez, F., & Font, X. (2007). Effects of thermal and mechanical pretreatments of secondary sludge on biogas production under thermophilic conditions, *Chemical Engineering Journal*, 133(1), 335–342.
24. Bougrier, C., Delgenes, J.-P., & Carrere, H. (2006). Combination of thermal treatments and anaerobic digestion to reduce sewage sludge quantity and improve biogas yield, *Process Safety and Environmental Protection*, 84(4), 280–284.
25. Bougrier, C., Delgenes, J.P., & Carrère, H. (2008). Effects of thermal treatments on five different waste activated sludge samples solubilisation, physical properties and anaerobic digestion, *Chemical Engineering Journal*, 139(2), 236–244.
26. Bougrier, C., Delgenes, J., & Carrere, H. (2007). Impacts of thermal pre-treatments on the semi-continuous anaerobic digestion of waste activated sludge, *Biochemical Engineering Journal*, 34(1), 20–27.
27. Bougrier, C., Albasi, C., Delgenès, J.P., & Carrère, H. (2006). Effect of ultrasonic, thermal and ozone pre-treatments on waste activated sludge solubilisation and anaerobic biodegradability, *Chemical Engineering and Processing: Process Intensification*, 45(8), 711–718.
28. Pilli, S., Bhunia, P., Yan, S., LeBlanc, R.J., Tyagi, R.D., & Surampalli, R.Y. (2011). Ultrasonic pretreatment of sludge: A review, *Ultrasonics Sonochemistry*, 18(1), 1–18.
29. Dhar, B.R., Nakhla, G., & Ray, M.B. (2012). Techno-economic evaluation of ultrasound and thermal pretreatments for enhanced anaerobic digestion of municipal waste activated sludge, *Waste Management*, 32(3), 542–549.
30. Chang, T.-C., You, S.J., Damodar, R.A., & Chen, Y.Y. (2011). Ultrasound pre-treatment step for performance enhancement in an aerobic sludge digestion process, *Journal of the Taiwan Institute of Chemical Engineers*, 42(5), 801–808.
31. Apul, O.G., & Sanin, F.D. (2010). Ultrasonic pretreatment and subsequent anaerobic digestion under different operational conditions, *Biore-source Technology*, 101(23), 8984–8992.
32. Park, W., & Ahn, J. (2011). Optimization of microwave pretreatment conditions to maximize methane production and methane yield in mesophilic anaerobic sludge digestion, *Environmental Technology*, 32(13), 1533–1540.
33. Karimi, K., & Taherzadeh, M.J. (2016). A critical review of analytical methods in pretreatment of lignocelluloses: Composition, imaging, and crystallinity, *Biore-source Technology*, 200, 1008–1018.
34. Safari, A., Karimi, K., & Shafiei, M. (2017). Dilute alkali pretreatment of softwood pine: A biorefinery approach, *Biore-source Technology*, 234, 67–76.
35. Liu, X., Wang, W., Gao, X., Zhou, Y., & Shen, R. (2012). Effect of thermal pretreatment on the physical and chemical properties of municipal biomass waste, *Waste Management*, 32(2), 249–255.
36. American Public Health Association, et al. (1915). Standard methods for the examination of water and wastewater. (Volume2), American Public Health Association, USA.
37. Kutz, M. (2013). Handbook of farm, dairy and food machinery engineering, Academic Press, USA.